

PEORIA AGDM, AB, Canada

WMO: 713480

Lat: 55.6211N

Lon: 118.2931W

Elev: 2038

StdP: 13.65

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-28.0	-21.6	-34.9	0.8	-28.0	-27.7	1.3	-21.3	22.1	33.1	20.0	29.8	3.0	80	0.671

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	81.6	61.3	78.4	59.9	75.4	58.7	64.8	76.4	62.8	74.1	60.8	72.0	7.8	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.5	84.8	69.5	58.3	78.4	67.1	56.3	72.7	64.7	30.9	77.1	29.3	74.1	27.9	71.7	73.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.0	16.7	14.6		-34.8	87.1	6.3	3.0	-39.3	89.3	-43.0	91.0	-46.5	92.7	-51.1	94.8
			DB	-34.7	68.8	6.2	2.8	-39.1	70.8	-42.8	72.5	-46.2	74.1	-50.7	76.2
			WB												

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.1	13.7	13.4	21.9	37.4	50.0	57.2	60.2	57.5	49.2	38.0	22.3	11.4
	DBStd	20.95	15.59	15.69	13.33	9.41	7.93	5.23	4.78	5.82	7.51	8.92	12.34	14.53
	HDD50	6030	1126	1024	871	384	96	6	1	9	103	380	833	1197
	HDD65	10558	1591	1444	1336	828	464	239	160	240	475	837	1282	1662
	CDD50	972	0	0	0	6	97	222	318	241	79	8	1	0
	CDD65	24	0	0	0	0	0	5	12	7	0	0	0	0
	CDH74	948	0	0	0	4	71	164	377	253	78	1	0	0
	CDH80	163	0	0	0	0	9	19	73	51	11	0	0	0
(14) Wind	WSAvg	6.4	6.0	5.7	6.2	7.5	8.0	6.9	6.0	5.7	6.5	6.6	6.0	5.7
(15) Precipitation	PrecAvg	18.5	1.3	0.8	0.9	1.0	1.8	3.0	2.9	2.1	1.5	1.2	1.2	0.9
	PrecMax	25.3	4.0	2.0	1.8	1.9	3.5	5.4	5.2	5.5	4.3	3.5	3.6	3.8
	PrecMin	12.1	0.4	0.2	0.4	0.2	0.7	0.7	1.1	0.7	0.3	0.2	0.2	0.0
	PrecStd	3.6	0.7	0.4	0.4	0.5	0.8	1.4	1.1	1.1	0.8	0.7	0.7	0.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	46.3	45.4	50.0	71.2	80.9	82.6	86.3	85.4	81.5	70.2	54.4	45.2
		MCWB	38.5	38.2	40.4	50.2	55.0	62.4	65.3	63.1	59.1	53.2	43.3	38.4
	2%	DB	43.1	41.6	46.3	62.9	75.1	78.4	81.6	80.4	75.7	60.8	47.1	41.4
		MCWB	37.3	35.7	38.3	46.0	53.4	58.8	62.6	61.9	57.8	48.1	39.9	35.6
	5%	DB	40.5	38.5	43.2	57.3	71.0	75.2	78.2	76.3	69.7	56.1	42.6	37.3
		MCWB	35.3	33.6	36.2	43.5	51.6	57.4	61.5	60.1	54.7	45.2	36.9	32.5
	10%	DB	36.3	35.0	39.8	52.8	67.3	71.7	74.9	72.8	64.4	52.1	38.8	32.9
		MCWB	32.3	31.2	34.1	41.1	49.8	56.2	60.2	58.6	52.0	43.1	34.3	29.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.3	38.7	41.6	51.5	58.0	64.9	68.8	66.4	61.6	54.1	43.9	38.8
		MCDB	44.9	44.1	48.4	69.5	75.3	78.2	79.8	79.2	76.6	68.2	51.2	44.3
	2%	WB	37.5	36.1	38.9	47.2	54.8	61.8	65.7	64.1	58.5	49.5	40.6	35.9
		MCDB	42.6	40.8	45.7	61.0	71.1	74.0	76.3	75.4	72.4	59.2	46.7	41.1
	5%	WB	35.5	33.9	36.5	44.1	52.8	59.5	63.5	62.1	55.8	46.3	37.4	32.8
		MCDB	40.2	38.1	42.4	55.5	67.8	71.1	74.6	73.2	68.0	54.2	42.2	37.0
	10%	WB	32.6	31.1	34.4	41.8	50.9	57.5	61.5	59.8	53.2	43.7	34.5	29.5
		MCDB	36.4	35.1	39.3	51.7	64.5	68.4	72.2	70.1	62.3	50.8	38.2	32.9
(35) Mean Daily Temperature Range	5% DB	MDBR	18.2	20.4	21.3	21.4	25.1	23.1	24.9	25.7	24.2	19.7	17.3	17.5
		MCDBR	20.4	21.1	22.8	30.4	32.6	30.9	32.5	33.0	33.6	27.4	19.8	22.7
		MCWBR	15.5	15.8	16.1	16.6	15.3	14.8	16.5	16.7	16.9	16.3	14.3	18.3
	5% WB	MCDBR	19.8	20.3	22.1	28.0	29.5	26.4	29.0	29.6	32.2	25.9	19.5	22.3
		MCWBR	15.3	15.5	15.8	16.0	14.3	14.0	16.1	16.2	16.7	16.3	14.6	18.2
(40) Clear-Sky Solar Irradiance	taub		0.244	0.249	0.303	0.326	0.344	0.352	0.353	0.353	0.310	0.285	0.244	0.231
	taud		2.278	2.356	2.275	2.377	2.392	2.389	2.406	2.396	2.523	2.548	2.417	2.228
	Ebn at Noon		223	271	275	285	285	283	280	272	273	255	226	197
	Edh at Noon		16	22	31	33	35	35	34	32	25	19	14	13
(44) All-Sky Solar Radiation	RadAvg		232	534	1009	1492	1780	1936	1885	1532	1045	559	252	163
	RadStd		24	42	80	85	122	150	119	133	91	64	22	14

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page